

HUTCHINSON COUNTY AP, TX, USA

WMO: 723635

Lat: 35.695N

Lon: 101.395W

Elev: 931

StdP: 90.63

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 03024

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-7.8	-18.0	0.9	0.6	-16.2	1.0	2.0	12.8	10.4	11.5	11.2	4.0	10	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	38.2	19.5	37.1	19.7	35.8	19.6	22.6	31.6	21.9	31.1	21.3	30.7	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	16.6	25.5	19.3	15.7	25.1	18.7	15.2	24.7	71.2	31.5	68.5	31.0	66.3	30.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	10.4	9.1	-15.1	40.8	2.3	1.6	-16.8	42.0	-18.1	43.0	-19.4	43.9	-21.1	45.1
			-15.4	23.8	2.7	1.6	-17.3	24.9	-18.9	25.8	-20.4	26.7	-22.4	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	4.4	6.0	10.9	15.5	20.4	25.8	28.1	27.2	23.0	16.3	10.0	4.5
	DBStd	9.80	5.56	6.24	5.71	5.25	4.93	3.52	2.90	3.13	3.97	5.11	5.55	5.53
	HDD10.0	658	186	139	58	14	1	0	0	0	0	11	67	183
	HDD18.3	1953	431	347	238	113	36	1	0	0	8	98	252	429
	CDD10.0	2869	13	26	86	178	323	474	561	533	390	205	67	12
	CDD18.3	1123	0	1	7	28	100	224	303	275	149	34	3	0
	CDH23.3	13667	2	21	126	421	1223	2671	3904	3279	1567	396	57	1
	CDH26.7	7082	0	4	33	155	568	1426	2233	1791	726	139	7	0
Wind	WSAvg	4.6	4.3	4.6	4.9	5.5	5.0	5.1	4.5	4.2	4.5	4.4	4.4	4.3
Precipitation	PrecAvg	524	15	16	24	47	77	69	71	74	44	37	22	21
	PrecMax	1048	47	42	109	216	254	152	221	137	118	132	120	88
	PrecMin	288	0	0	0	0	12	2	9	13	0	0	0	3
	PrecStd	143	13	11	25	47	48	41	43	33	30	37	26	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	26.1	30.1	33.2	37.1	40.3	40.4	39.1	37.3	33.2	27.7	22.3
		MCWB	9.4	10.3	12.6	13.7	16.5	17.4	19.6	19.9	18.7	15.9	12.3	9.3
	2%	DB	19.8	22.5	27.0	30.9	34.6	37.5	38.3	37.6	35.0	30.3	25.0	18.7
		MCWB	8.0	9.4	12.0	13.3	15.7	18.8	20.0	20.1	18.5	15.9	11.7	8.2
	5%	DB	16.9	19.8	23.8	28.0	32.5	35.7	37.1	36.3	33.0	27.8	22.3	16.2
		MCWB	7.0	8.5	10.8	12.9	16.2	19.3	20.2	20.2	18.7	15.2	11.0	7.1
	10%	DB	13.8	16.5	21.1	25.5	30.1	33.7	35.6	34.7	31.1	25.1	19.1	13.6
		MCWB	5.5	6.7	10.2	12.6	16.0	19.5	20.2	20.1	18.5	14.5	9.7	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	12.4	15.9	18.0	20.8	23.4	23.8	23.5	21.8	20.2	15.3	11.6
		MCDB	19.7	21.9	22.9	24.2	28.8	31.9	32.6	31.6	30.4	26.2	21.7	17.4
	2%	WB	8.9	10.2	14.2	16.4	19.8	22.2	22.6	22.4	20.9	18.6	13.6	9.2
		MCDB	18.1	20.4	20.4	23.2	27.7	31.2	31.7	31.4	29.3	24.5	19.9	17.0
	5%	WB	7.4	8.9	12.6	15.1	18.9	21.5	21.8	21.7	20.2	17.5	12.3	7.7
		MCDB	16.3	18.8	20.7	22.6	27.2	30.6	31.5	30.9	28.5	23.2	19.2	15.1
	10%	WB	5.8	7.5	11.0	13.9	17.9	20.8	21.3	21.1	19.6	16.1	10.8	6.2
		MCDB	13.5	16.0	20.0	22.5	25.9	29.9	31.2	30.5	27.7	22.1	18.7	13.3
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.0	14.3	14.1	13.7	13.3	12.8	12.7	12.8	13.2	13.6	12.5
		MCDBR	17.6	18.7	18.1	18.0	16.6	15.6	14.5	14.4	14.9	16.4	17.2	16.5
		MCWBR	9.3	9.2	7.9	7.2	6.1	4.8	3.8	3.9	4.5	6.5	7.9	8.8
	5% WB	MCDBR	16.7	17.4	15.4	14.4	14.1	13.5	12.8	12.8	12.3	12.5	14.4	15.4
		MCWBR	9.0	9.1	7.6	7.4	6.4	5.0	4.1	4.1	4.4	6.4	7.4	8.7
Clear-Sky Solar Irradiance	taub	0.257	0.265	0.295	0.328	0.343	0.380	0.384	0.373	0.343	0.302	0.275	0.258	
	taud	2.596	2.585	2.501	2.419	2.437	2.380	2.405	2.404	2.467	2.566	2.566	2.592	
	Ebn at Noon	967	993	980	955	939	899	893	899	915	934	933	943	
	Edh at Noon	73	84	100	115	114	121	117	115	103	85	75	69	
All-Sky Solar Radiation	RadAvg	3.04	3.85	5.08	6.29	7.01	7.48	7.25	6.46	5.53	4.17	3.24	2.63	
	RadStd	0.27	0.36	0.37	0.37	0.53	0.46	0.31	0.32	0.27	0.47	0.32	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+98	N/A

Nomenclature: See separate page